Xinyang(Oliver) Zhou

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EDUCATION

Northwestern University **Evanston, IL**
M.S. in Machine Learning and Data Science | GPA: 3.94/4.0 Sep 2023 - Dec 2024
Relevant Coursework: Generative AI, NLP, Deep Learning, Big Data Analytics, Cloud Engineering for Data Science

University of Michigan **Ann Arbor, MI**
B.S. in Statistics, Minor in Mathematics and Business | GPA: 3.84/4.0 Aug 2019 - May 2023

TECHNICAL SKILLS *(GitHub Link)*

Programming: Python, R, SQL, C++ **ML Toolkits**: PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformer
Infrastructure: AWS, Snowflake, GitHub, Docker, Airflow, Hadoop, Spark, Hive
Analytics and Visualization: Tableau, Power BI, Plotly, ggplot2, Matplotlib, Seaborn

WORK EXPERIENCE

Freddie Mac **McLean, VA**
Quantitative Analyst Apr 2025 – Present

- Designed and deployed an end-to-end loan prospector model, integrating data from upstream model outputs and internal systems and applying **imputation** methods to address missing values. Achieved 0.975 accuracy and 0.92 F1 score on a balanced testing set, effectively capturing default risk behavior to support business decision
- Implemented automated model monitoring to track data quality, class balance, and drift, cutting annual losses by ~$2 M through proactive performance control
- Performed **model validation** (subpopulations, out-of-time testing, sensitivity analysis, k-fold cross-validation) to ensure robustness and regulatory compliance
- Designed and implemented custom model explainability metrics (**Gini Coefficient, Tail-10%**), and presented **audit-ready reports** to both technical reviewers and business stakeholders

Amazon **Chicago, IL**
Data Science Intern - Capstone Sep 2024 – Dec 2024

- Applied **NLP** and **LSTM** sequence modeling on system fault logs to capture contextual and temporal patterns, automating root-cause matching and improving diagnostic speed by 15% utilizing vector database
- Built a **logistic regression** model to quantify relationships between structured system information (e.g., loading time, latency) and fault severity, achieving an AUC=0.87 and enabling proactive anomaly classification

CDK Global **Chicago, IL**
Data Science & Software Engineer Intern Jun 2024 – Aug 2024

- Developed an **unsupervised ML algorithm** to generate customer propensity scores, enhancing campaign targeting accuracy by 22%, providing potential revenue growth by ~150K annually
- Built a pipeline using **XGBoost** and **Difference-in-Differences** to analyze new customer promotion strategy, identifying segments and driving a measurable uplift in signup rates
- Engineered data pipelines in **SQL + Snowflake** to correct mapping errors and monitor quality metrics, protecting marketing accuracy and driving a 10% retention rate increase
- Streamlined and synced ML models via **Docker** and **GitHub**, ensuring cross-team collaboration efficiency and code reproducibility

Nationwide Mutual Insurance Company **Columbus, OH**
Business Insights Analyst Intern May 2023 – Aug 2023

- Designed **A/B experiments** to measure service performance across 41 U.S. states, improving regional accuracy and boosting productivity by 15%
- Built **interactive dashboards (Tableau + Power BI)** to inform pricing strategy, contributing to a 3.7% revenue increase and mitigating high-risk exposure to avoid financial stress
- Authored advanced SQL pipelines (CTEs, window functions) to automate reporting and downstream analytics

RESEARCH EXPERIENCE

Harvard University, Harvard Business School
Research Assistant / Co-Author

- Designed a modular ML pipeline for tabular data processing, SMOTE for class imbalance, covariate selection, and prediction generation using **LLM APIs** with **prompt engineering** strategies (e.g., few-shot)
- Enhanced large language model performance through **fine-tuning** and prompt optimization. Demonstrating 20% reliability improvement compared to the baseline prediction
- Defined and tracked **custom business metrics** to evaluate LLM reliability compared to traditional ML models; applied **causal inference methods** (e.g., average treatment effect, statistical significance testing) to uncover robust insights